

JEDEC STANDARD

High Temperature Storage Life

JESD22-A103D

(Revision of JESD22-A103C, November 2004)

DECEMBER 2010

JEDEC SOLID STATE TECHNOLOGY ASSOCIATION



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TEST METHOD A103D**HIGH TEMPERATURE STORAGE LIFE**

(From JEDEC Board Ballot JCB-10-58 and JCB-10-65, formulated under the cognizance of JC-14.1 Subcommittee on Reliability Test Methods for Packaged Devices.)

1 Scope

The test is applicable for evaluation, screening, monitoring, and/or qualification of all solid state devices.

The high temperature storage test is typically used to determine the effects of time and temperature, under storage conditions, for thermally activated failure mechanisms and time-to-failure distributions of solid state electronic devices, including nonvolatile memory devices (data retention failure mechanisms). Thermally activated failure mechanisms are modeled using the Arrhenius Equation for acceleration. During the test, accelerated stress temperatures are used without electrical conditions applied. This test may be destructive, depending on time, temperature and packaging (if any).

2 Reference documents (informative)

JESD22-B101, *External Visual*.

JESD47, *Stress-Test-Driven Qualification of Integrated Circuits*.

J-STD-020, Joint IPC/JEDEC Standard, *Moisture/Reflow Sensitivity Classification for Nonhermetic Solid State Surface-Mount Devices*.

3 Apparatus

3.1 High temperature storage chambers

The apparatus required for this test shall consist of a controlled temperature chamber capable of maintaining the specified temperature over the entire sample population under test.

3.2 Electrical test equipment

Electrical equipment capable of performing the appropriate measurements for the devices being tested, including write and verify the required data retention pattern(s) for nonvolatile memories.

4 Procedure

4.1 High temperature storage conditions

The devices under test shall be subjected to continuous storage at one of the temperature conditions of Table 1.

Table 1 — High temperature storage conditions

Condition A: +125 (-0/+10) °C
Condition B: +150 (-0/+10) °C
Condition C: +175 (-0/+10) °C
Condition D: +200 (-0/+10) °C
Condition E: +250 (-0/+10) °C
Condition F: +300 (-0/+10) °C

NOTE CAUTION should be exercised when selecting an accelerated test condition since the accelerated temperature used may exceed the capabilities of the device and materials, thereby inducing (overstress) failures that would not occur under normal use conditions.

As a minimum the following items should be taken into consideration:

- 1) Melting point of metals present, especially solder. Degradation of metals including metallurgical interfaces.
- 2) Package degradation. For example: glass transition temperature and thermal stability (in air) of any polymeric materials.
- 3) Moisture rating of package (per J-STD-020).
- 4) Temperature limitations of silicon devices. For example: charge loss in nonvolatile memories.
- 5) Test conditions (temperature, time) should be selected to cover the acceleration of the corresponding failure mechanism and the anticipated lifetime (operational time) of the device.

Qualification (JESD47) and reliability monitoring test conditions typically require a duration of 1000 hours per condition B of Table 1. Other conditions and durations may be used as appropriate. JESD47¹ also recommends that some package styles be subjected to SMT reflow simulation prior to stress.

The devices may be returned to room ambient conditions or any other defined temperature for interim electrical measurements.

¹ From Table 2 of JESD47 - Preconditioning to JESD22A113 is recommended, specifically for wirebonded products qualified to Pb-free reflow profiles. Moisture soak as part of the preconditioning is optional.

4 Procedure (cont'd)

4.2 Measurements

Unless otherwise specified, interim and final electrical test measurements shall be completed within 168 hours after removal of the devices from the specified test conditions. Interim measurements are optional unless otherwise specified. The time window need not be met if verification data for a given technology is provided. If the final readpoint time window is exceeded then the units may be restressed for the same amount of time that the window is exceeded.

The electrical test measurements shall consist of parametric and functional tests specified in the applicable procurement document. For nonvolatile memories, the data specified data retention pattern must be written initially, and then subsequently verified without re-writing.

4.3 Failure criteria

A device will be considered a high temperature storage failure if parametric limits are exceeded, or if functionality cannot be demonstrated under nominal and worst-case conditions, as specified in the applicable procurement document. For nonvolatile memories, the specified data retention pattern shall be verified before and after storage. A margin test may be used to detect data retention degradation.

Mechanical damage, such as cracking, chipping, or breaking of the package, (as defined in test method B101 "External Visual") will be considered a failure, provided that such damage was not induced by fixtures or handling and it is critical to the package performance in the specific application.

Cosmetic package defects and degradation of lead finish, or solderability are not considered valid failure criteria for this stress.

5 Summary

The following details shall be specified in the applicable procurement document.

- a) Electrical test measurements, failure criteria and specifications
- b) Sample size and number of failures (specify zero if none observed).
- c) Time and conditions, if other than 1000 hours per condition B.
- d) Interim electrical test measurements, if required.
- e) Nonvolatile memory data retention pattern (for appropriate devices)

Annex A (informative) Differences between JESD22-A103D and JESD22-A103C

This table briefly describes most of the changes made to entries that appear in this standard, JESD22-A103D, compared to its predecessor, JESD22-A103C (November 2004). If the change to a concept involves any words added or deleted (excluding deletion of accidentally repeated words), it is included. Some punctuation changes are not included.

Clause	Description of change
1	In Scope, added reference that thermally activated failure mechanisms are Arrhenius based.
2	New section (2) added to denote documents referenced by this test method Renumbered clauses accordingly based on new clause added.
4.1 (5)	Added reference from JESD47 that recommends preconditioning for some package types prior to stress
4.2	First paragraph, modified time window requirement from 96 hours to 168 hours
4.2	First paragraph, add last 2 sentences



Standard Improvement Form**JEDEC JESD22-A103D**

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1. I recommend changes to the following:

☐ Requirement, clause number _____

☐ Test method number _____ Clause number _____

The referenced clause number has proven to be:

☐ Unclear ☐ Too Rigid ☐ In Error

☐ Other _____

2. Recommendations for correction:

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